

Hot springs and geysers

In some volcanic regions, water seeping into the ground comes into contact with very hot rock. It usually bubbles back to the surface as a hot spring, but it can also erupt as an explosive geyser. In the deep ocean, the same process creates “black smokers” of superheated water.

BOILING UP

Hot springs are found in highly volcanic regions where molten rock lies near the surface. High pressure underground heats water to well above its normal boiling point. It is then forced to the surface, where it forms steaming hot pools. These are often vividly colored by microbes called archaea that can survive in extreme conditions.



► WARM GLOW

Dense colonies of microscopic life create a dazzling effect in the Morning Glory Pool, a hot spring in Yellowstone National Park.

DISSOLVED MINERALS

Water that is superheated at high pressure deep below the surface can reach a temperature of 572°F (300°C) or more. This allows it to dissolve minerals from the rocks, but the minerals turn solid again when the water reaches the surface and its temperature drops. This process can result in spectacular terraced deposits of minerals building up in places where the water flows away from the hot spring.

▼ TRAVERTINE TERRACES

Carbonate minerals known as travertine hang like frozen waterfalls from the terraced pools around the Pamukkale hot springs in southwestern Turkey.



MUD AND GAS

The hot water and gas that rise to the surface often react with the rocks, turning them to clay. The mixture of clay and water forms pools of hot liquid mud that pop and bubble with rising gas. Other vents called fumaroles release clouds of steam that often reek of sulfurous volcanic gases—the “rotten egg” smell.

